Novel amplicons on the short arm of chromosome 7 identified using high resolution array CGH contain over expressed genes in addition to EGFR in glioblastoma multiforme.

PMID: 16110500 | Indexed in PubMed

Amplification of a defined chromosome segment on the short arm of chromosome 7 has frequently been reported in glioblastoma multiforme (GBM), where it is generally assumed that it is the result of over expression of the epidermal growth factor receptor (EGFR) gene that provides the selective pressure to maintain the amplification event. We have used high resolution array comparative genomic hybridization (aCGH) to analyze amplification events on chromosome 7p in GBM, which demonstrates that, in fact, several other regions distinct from EGFR can be amplified. To determine the changes in gene expression levels associated with these amplification events, we used oligonucleotide expression arrays to investigate which of the genes in the amplified regions were also over expressed. These analyses demonstrated that not all genes in the amplicons showed increased expression, and we have defined a series of over expressed genes on 7p that could potentially contribute to the development of the malignant phenotype in these tumors. The global analysis of amplification afforded by aCGH analysis has improved our ability to define numerical chromosome abnormalities in cancer cells and has raised the possibility that genes other than EGFR may be important.

Chromosome 7 in glioblastoma tissue. Parenchymal vs. endothelial cells.

PMID: 7497447 | Indexed in PubMed

Strong endothelial proliferation is a prominent feature of glioblastomas and sometimes these proliferated areas transform into a malignant component of glioblastoma, resulting in gliosarcomas. It has not been established whether the proliferated endothelial areas are cytogenetically abnormal. To clarify this question, the most common cytogenetic aberration, gain of chromosome 7, was chosen and in situ hybridization was performed on paraffin-embedded tissue sections of three glioblastomas. The purpose was to compare the parenchymal tumor cells and the endothelial cells. The results showed trisomy 7 in only a small amount of endothelial cells (5-8%), whereas 23-38% of parenchymal tumor cells displayed trisomy 7.

Frequent mitotic errors in tumor cells of genetically micro-heterogeneous glioblastomas.

PMID: 11701945 | Indexed in PubMed

Glioblastoma multiforme (GBM) is characterized by intratumoral heterogeneity as to both histomorphology and genetic changes, displaying a wide variety of numerical chromosome aberrations the most common of which are monosomy 10 and trisomy 7. Moreover, GBM in vitro are known to have variable karyotypes within a given tumor cell culture leading to rapid karyotype evolution through a high incidence of secondary numerical chromosome aberrations. The aim of our study was to investigate to what extent this mitotic instability of glioblastoma cells is also present in vivo. We assessed the spatial distribution patterns of numerical chromosome aberrations in vivo in a series of 24 GBM using two-color in situ hybridization for chromosomes 7/10, 8/17, and 12/18 on consecutive 6-microm paraffin-embedded tissue slides. The chromosome aberration patterns were compared with the histomorphology of the investigated tumor assessed from a consecutive HE-stained section, and with the in vitro karyotype of cell cultures established from the tumors. All investigated chromosomes showed mitotic instability, i.e., numerical aberrations within significant amounts of tumor cells in a scattered distribution through the tumor tissue. As to chromosomes 10 and 17, only monosomy occurred, as to chromosome 7 only trisomy/polysomy, apparently as a result of selection in favor of the respective aberration. Conversely, chromosomes 8, 12, and 18 displayed scattered patterns of monosomy as well as trisomy within a given tumor reflecting a high mitotic error rate without selective effects. The karyotypes of the tumor cell cultures showed less variability of numerical aberrations apparently due to clonal adaptation to in vitro conditions. We conclude that glioblastoma cells in vivo are characterized by an extensive tendency to mitotic errors. The resulting clonal diversity of chromosomally aberrant cells may be an important biological constituent of the well-known ability of glioblastomas to preserve viable tumor cell clones under adaptive stress in vivo, in clinical terms to rapidly recur after antitumoral therapy including radio- or chemotherapy.

Genetic analysis of human glioblastomas using a genomic microarray system.

PMID: 15696966 | Indexed in PubMed

Genomic microarray systems can simultaneously provide substantial genetic and chromosomal information in a relatively short time. We have analyzed genomic DNA from frozen sections of 30 cases of primary glioblastomas by GenoSensor Array 300 in order to characterize gene amplifications, gene deletions, and chromosomal information in the whole genome. Genes that were frequently amplified included RFC2/CYLN2 (63.3%), EGFR (53.3%), IL6 (53.3%), ABCB1 (MDR1) (36.7%), and PDGFRA (26.7%). Genes that were frequently deleted included (56.7%), FGFR2 (66.7%), MTAP (60.0%), DMBT1 CDKN2A (p16)/MTAP (50.0%), PIK3CA (43.3%), and EGR2 (43.3%), but deletion of RB1 or TP53 was rarely detected. Chromosomal gains were observed frequently for 7q (33.3%), 7p (20.0%), and 17q (13.3%). Loss of the 10q was frequently detected in 13 of 30 cases (46.7%). Loss of the entire chromosome 10 was seen in 9 of 30 cases (30.0%), and was often accompanied by EGFR amplification (7 cases, 77.8%). The GenoSensor Array 300 proved to be useful for identification of genome-wide molecular changes in glioblastomas. The obtained microarray profile can also yield valuable insight into the molecular events underlying carcinogenesis of brain tumors and may provide clues about clinical correlations, including response to treatment.

MEOX2 homeobox gene promotes growth of malignant gliomas.

PMID: 35468210 | Indexed in PubMed

Glioblastoma (GBM) is an aggressive tumor that frequently exhibits gain of chromosome 7, loss of chromosome 10, and aberrantly activated receptor tyrosine kinase signaling pathways. Previously, we identified Mesenchyme Homeobox 2 (MEOX2), a gene located on chromosome 7, as an upregulated transcription factor in GBM. Overexpressed transcription factors can be involved in driving GBM. Here, we aimed to address the role of MEOX2 in GBM. Patient-derived GBM tumorspheres were used to constitutively knockdown or overexpress MEOX2 and subjected to in vitro assays including western blot to assess ERK phosphorylation. Cerebral organoid models were used to investigate the role of MEOX2 in growth initiation. Intracranial mouse implantation models were used to assess the tumorigenic potential of MEOX2. RNA-sequencing, ACT-seq, and CUT&Tag were used to identify MEOX2 target genes. MEOX2 enhanced ERK signaling through a feed-forward mechanism. We identified Ser155 as a putative ERK-dependent phosphorylation site upstream of the homeobox-domain of MEOX2. S155A substitution had a major effect on MEOX2 protein levels and altered its subnuclear localization. MEOX2 overexpression cooperated with p53 and PTEN loss in cerebral organoid models of human malignant gliomas to induce cell proliferation. Using high-throughput genomics, we identified putative transcriptional target genes of MEOX2 in patient-derived GBM tumorsphere models and a fresh frozen GBM tumor. We identified MEOX2 as an oncogenic transcription regulator in GBM. MEOX2 increases proliferation in cerebral organoid models of GBM and feeds into ERK signaling that represents a core signaling pathway in GBM.

Interphase cytogenetic (in situ hybridization) analysis of astrocytomas using archival, formalin-fixed, paraffin-embedded tissue and nonfluorescent light microscopy.

PMID: 9260757 | Indexed in PubMed

Astrocytomas contain nonrandom chromosomal abnormalities that recently have been correlated with shortened patient survival. Two frequently reported aberrations are trisomy 7 and monosomy 10. We assessed the numerical complement of chromosomes 7 and 10 in formalin-fixed, paraffin-embedded brain biopsy tissue from 28 diffuse astrocytomas by in situ hybridization using a nonfluorescent enzymatic detection system. Clinical follow-up of at least 5 years was available in 26 cases (93%). Monosomy 10 was identified in 7 cases (25%): astrocytoma, 1 case; anaplastic astrocytoma, 1 case; and glioblastoma, 5 cases. Trisomy 7 was identified in 11 cases (39%): astrocytoma, 5 cases; glioblastoma, 6 cases. Multivariate analysis revealed that monosomy 10 was the most statistically significant negative predictor of patient survival. Numerical chromosomal abnormalities are detectable in astrocytomas in archival tissue using interphase cytogenetics and nonfluorescent light microscopy. Although larger studies are required, our data suggest that potentially useful prognostic information may be obtained with this approach.

Glioblastomatous recurrence of oligodendroglioma remote from the original site: a case report.

PMID: 17145331 | Indexed in PubMed

As in all diffuse gliomas, recurrence is an inherent feature of oligodendrogliomas, either as the same or higher grade neoplasm at the primary site. The rate of remote recurrence after surgery for the primary tumor cannot be estimated from the scarce literature, but delayed treatment of the primary tumor and genetic alterations may be associated with this phenomenon. A 40-year-old man presented with generalized seizures. A magnetic resonance imaging scan disclosed a right frontal mass lesion showing features of a low-grade glioma for which he refused any treatment. Seven months after diagnosis upon uncontrollable seizures, he underwent a stereotactic biopsy, which was followed by a right frontal craniotomy, both of which confirmed the lesion as a grade 2 oligodendroglioma. Six months after surgery, the patient presented with a left frontal lobe GBM without evidence of recurrence at the primary site. The genetic analysis of the primary and recurrent tumors showed trisomy 7, monosomy 10, but not 1p or 19q deletions, which have been proposed as markers for favorable prognosis. Recurrence of a frontal lobe oligodendroglioma remote from the primary site as a GBM is a rare occurrence. Single-cell invasion across the corpus callosum with subsequent or simultaneous malignant degeneration into a secondary GBM is the likely mechanism. As the genetic analysis suggests, conversion of oligodendroglioma to GBM may be associated with gain of chromosome 7, loss of chromosome 10, and other genetic markers that may represent late events in the oncogenesis of oligodendroglial tumors.

Screening of genomic imbalances in glioblastoma multiforme using high-resolution comparative genomic hybridization.

PMID: 17203188 | Indexed in PubMed

Comparative genomic hybridization (CGH) is a molecular cytogenetic technique that allows the genome-wide analysis of DNA sequence copy number differences. We applied conventional CGH and the recently developed high-resolution CGH (HR-CGH) to tumour samples from 18 patients with glioblastoma multiforme (GBM) in order to compare the sensitivity of CGH and HR-CGH in the screening of chromosomal abnormalities. The abnormalities were studied in topologically different central and peripheral tumour parts. A total of 78 different changes were observed using CGH (0-16 per tumour, median 3.5) and 154 using HR-CGH (0-21 per tumour, median 6). Using HR-CGH, losses were more frequent than gains. The representation of the most prominent changes revealed by both methods was similar and was comprised of the amplification of 7q12 and 12q13-q15, the gain of 7, 3q and 19, and the loss of 10, 9p, and 13q. However, HR-CGH detected certain other abnormalities (the loss of 6, 14q, 15q and 18q, and the gain of 19), which were rarely revealed by CGH. Using HR-CGH, the numbers and types of chromosomal changes detected in the central and peripheral parts of GBM were almost the same. The loss of chromosomes 10 and 9p and the gain of chromosomes 7 and 19 were the most frequent chromosomal alterations in both tumour parts. Our results from the GBM analysis show that HR-CGH technology can reveal new, recurrent genetic alterations involving the genes known to participate in tumorigenesis and in the progression of several human malignancies, thus allowing for a more accurate genetic characterization of these tumours.

Association of chromosome 7, chromosome 10 and EGFR gene amplification in glioblastoma multiforme.

PMID: 16167544 | Indexed in PubMed

Glioblastoma multiforme (GBM) is characterized by intratumoral heterogeneity in both histomorphological and genetic changes, displaying a wide variety of numerical chromosome aberrations, the most common of which are trisomy 7 and monosomy 10. The amplification of the epidermal growth factor receptor (EGFR) gene is the most frequently reported genetic abnormality. The associations between these parameters and their implication in the tumoral progression are poorly understood. We performed simultaneous fluorescence in situ hybridization (FISH) with centromeric DNA probes for chromosomes 7 and 10 in smear preparations, and EGFR gene amplification by PCR from 25 cases of GBM. Trisomy/ polysomy for chromosome 7 was present in 76% of cases and monosomy 10 in 68%. Both alterations were associated in 56% of cases. The EGFR gene was amplified in 52% of tumors; in 44% associated with trisomy/ polysomy 7, and in 36% with monosomy 10. The three parameters were associated together in 28% of cases. Kaplan-Meier survival rate analysis demonstrated lower survival rates in patients with

monosomy 10, trisomy 7, and monosomy associated with trisomy 7. The other combinations were not different in frequency in relation to survival. In the present study, trisomy/polysomy 7 and monosomy 10 have been found to be frequently associated. The combination of both anomalies is probably important in the tumorigenesis of glioblastoma. Moreover, this association is apparently independent of EGFR gene amplification, which could be a later event in this process.

Comparative genetic patterns of glioblastoma multiforme: potential diagnostic tool for tumor classification.

PMID: 11302337 | Indexed in PubMed

Cytogenetic and molecular genetic studies of glioblastoma multiforme (GBM) have shown that the most frequent alterations are gains of chromosome 7, losses of 9p loci and chromosome 10, and gene amplification, primarily of the epidermal growth factor receptor (EGFR) gene. Although this profile is potentially useful in distinguishing GBM from other tumor types, the techniques used tend to be labor intensive, and some can detect only gains or losses of genetic loci. Comparative genomic hybridization (CGH) is a powerful technique capable of identifying both gains and losses of DNA sequences. The present study compares the CGH evaluation of 22 GBM with classic cytogenetics, loss of heterozygosity by allelotyping, and gene amplification by Southern blot analysis to determine the reliability of CGH in the genetic characterization of GBM. The CGH and karyotypic data were consistent in showing gain of chromosome 7 accompanied by a loss of chromosome 10 as the most frequent abnormality, followed by a loss of 9p in 17 of 22 GBM cases. Loss of heterozygosity of chromosomes 10 (19/22) and 9p (9/22) loci confirmed the underrepresentation by CGH. Genomic amplifications were observed by CGH in 5 of the 10 cases where gene amplification was detected by Southern blot analysis. The data show that CGH is equally reliable, compared with the more established genetic methods, for recognizing the prominent genetic alterations associated with GBM and support its use as a plausible adjunct to glioma classification.